Senior Software Engineer - Image Reconstruction (C++/CUDA)

Philips · Orange (OH), OH

Job ID (this archive)	HI-168
Employer	Philips
Location(s)	Orange (OH), OH
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Biomedical Engineering, Electrical Engineering, Computer Engineering, Computer Science, Software Engineering, Data Science
Pay listed	$114,750 – $183,600 per year
Employment type	Full time
Posted / listing date	2026-10-06
Employer requisition	580470
Recruiting status when checked	Open when checked (Oct 11, 2026). Employer note: Open — employer posting available Saved PDF = full employer posting text, reformatted for print.
Source page	https://philips.wd3.myworkdayjobs.com/en-US/jobs-and-careers/job/Orange-OH-Ohio-United-States/R-D-Software-Development-Engineer--Medical_580470
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> Archived job-description text captured 2026-10-11. PDF recreated October 11, 2026 from the previously saved description. Original wording retained; web layout and navigation omitted. Live listings change and close; check the source page before applying.

Job description

Job TitleSenior Software Engineer - Image Reconstruction (C++/CUDA)

Job Description

Build the GPU-native engine that turns raw CT physics into life-saving images inside Philips scanners worldwide, wringing maximum performance from constrained hardware at the frontier of C++, CUDA, and AI alongside world-class physicists.

Your role:

Help re-architect an entire CT image-reconstruction pipeline, from raw detector physics to the final clinical image, as a fully GPU-native, massively scalable platform in C++ and CUDA.

Work shoulder-to-shoulder with physicists, algorithm architects, and platform engineers, translating complex signal and image-processing models into high-performance GPU implementations that balance image quality against compute cost.

Collaborate with CT platform teams around the globe to design, build, test, and deploy an industry-leading reconstruction software platform.

Push the frontier of performance engineering and applied AI by profiling, parallelizing, optimizing memory use across caches and shared memory, and pioneering learned models that replace expensive physics with fast, accurate equivalents.

This hybrid role is based in Cleveland, OH, with three flexible days in the office and two remote days each week. Regular travel is not typically required.

You're the right fit if:

You have 5+ years of professional experience developing and maintaining complex software systems in C++. You have experience with GPU programming (CUDA, OpenCL, or HIP), or you have optimized code for resource-constrained or embedded environments and are eager to transition into GPU-centric development.

You bring a strong mathematical foundation, and a background in image and signal processing, including the theory of medical image formation, is a plus. You are energized by performance engineering, and enjoy profiling, parallelization, and squeezing every bit of throughput from modern GPUs.

You have a bachelor's degree or higher in Computer Science, Electrical or Computer Engineering, Mathematics, Physics, Biomedical Engineering, or a related field. Curiosity about applying machine learning to accelerate scientific or physics computation is a strong plus.

You have excellent written and verbal communication skills, able to explain complex designs, architectures, and algorithms to non-specialists. You have the ability and desire to collaborate closely with CT global development teams around the world.

You must be able to successfully perform the following minimum Physical, Cognitive and Environmental job requirements with or without accommodation for this position.

How we work together

We believe that we are better together than apart. For our office-based teams, this means working in-person at least 3 days per week. Onsite roles require full-time presence in the company's facilities. Field roles are most effectively done outside of the company's main facilities, generally at the customers' or suppliers' locations.

This is an office role.

About Philips

We are a health technology company. We built our entire company around the belief that every human matters, and we won't stop until everybody everywhere has access to the quality healthcare that we all deserve. Do the work of your life to help improve the lives of others.

Learn more about our business.

Discover our rich and exciting history.

Learn more about our purpose.

Learn more about our culture.

Philips Transparency Details

The pay range for this position in Orange, OH is $114,750 to $183,600.00.

The actual base pay offered may vary within the posted ranges depending on multiple factors including job-related knowledge/skills, experience, business needs, geographical location, and internal equity.

In addition, other compensation, such as an annual incentive bonus, sales commission or long-term incentives may be offered. Employees are eligible to participate in our comprehensive Philips Total Rewards benefits program, which includes a generous PTO, 401k (up to 7% match), HSA (with company contribution), stock purchase plan, education reimbursement and much more. Details about our benefits can be found here.

At Philips, it is not typical for an individual to be hired at or near the top end of the range for their role and compensation decisions are dependent upon the facts and circumstances of each case.

Additional Information

US work authorization is a precondition of employment. The company will not consider candidates who require sponsorship for a work-authorized visa, now or in the future.

Company relocation benefits will not be provided for this position. For this position, you must reside in or within commuting distance to Orange, Ohio.

#LI-PHI

This requisition is expected to stay active for 45 days but may close earlier if a successful candidate is selected or business necessity dictates. Interested candidates are encouraged to apply as soon as possible to ensure consideration.

Philips is an Equal Employment and Opportunity Employer including Disability/Vets and maintains a drug-free workplace.

Notes for ECS 1100 students

Course spreadsheet notes

Skills mentioned: C++; CUDA/OpenCL/HIP; image reconstruction; physics; optimization.

Software Engineering

Direct role in this area. See employer's specific degree disciplines.

Degree: Bachelor's or higher required

Experience / eligibility: 5+ years complex C++ development. GPU programming or optimization experience.

Other: No work-visa sponsorship. See full posting for local conditions.

Other majors named: CS, Electrical/Computer Engineering, Mathematics, Physics, BME or related

Career stage: Experienced / senior / Full-time / experienced

Salary evidence: The pay range for this position in Orange, OH is $114,750 to $183,600.00.